

Scenario 2 output 1 ollama

Here is the analysis of potential threats in the LLM-integrated secure file management system:

****STRIDE Threats****

1. ****T1: Unauthorized Data Access****

- * Category: Authorization
- * Affected component/data flow: File Management Service (P2), Server Storage Repository (DS1)
- * Attack description: An attacker exploits a vulnerability in P2 to access files without proper authorization.
- * Impact: Confidentiality, Integrity, and Availability of files are compromised.
- * Severity: High
- * Likelihood: Medium
- * Mitigation: Implement robust authentication and authorization mechanisms in P2.

2. ****T2: Data Tampering****

- * Category: File Security
- * Affected component/data flow: Server Storage Repository (DS1)
- * Attack description: An attacker modifies file contents or metadata, compromising the integrity of files.
- * Impact: Integrity of files is compromised.
- * Severity: Medium
- * Likelihood: Low
- * Mitigation: Implement data checksums and versioning in DS1.

3. ****T3: Elevation of Privilege****

- * Category: Authorization
- * Affected component/data flow: AI File Assistant (P3), LLM Integration Service (P4)
- * Attack description: P3 or P4 gains unauthorized access to sensitive data or resources.
- * Impact: Confidentiality, Integrity, and Availability of files are compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement strict access controls in P2 and limit permissions for P3 and P4.

****LLM-Specific Threats****

1. ****T4: LLM Model Poisoning****

- * Category: File Security
- * Affected component/data flow: External Entity EE2 (External LLM Provider)

- * Attack description: An attacker injects malicious data into the LLM model, compromising its accuracy or behavior.
- * Impact: Integrity of LLM output is compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement input validation and sanitization in P4.

****OWASP LLM Top 10 Threats****

1. **T5: Insecure Data Processing**

- * Category: File Security
- * Affected component/data flow: AI File Assistant (P3), Server Storage Repository (DS1)
- * Attack description: An attacker exploits vulnerabilities in data processing, compromising the integrity of files.
- * Impact: Integrity of files is compromised.
- * Severity: Medium
- * Likelihood: Low
- * Mitigation: Implement robust error handling and input validation in P3.

****Authorization Threats****

1. **T6: Broken Access Control**

- * Category: Authorization
- * Affected component/data flow: File Management Service (P2), Server Storage Repository (DS1)
- * Attack description: An attacker exploits vulnerabilities in access control mechanisms, compromising file access.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Medium
- * Mitigation: Implement robust access controls in P2.

****Data Leakage Threats****

1. **T7: Data Exposure**

- * Category: File Security
- * Affected component/data flow: Server Storage Repository (DS1)
- * Attack description: An attacker accesses sensitive data without proper authorization.
- * Impact: Confidentiality of files is compromised.
- * Severity: High
- * Likelihood: Medium
- * Mitigation: Implement robust encryption and access controls in DS1.

****LLM-Specific Threats****

1. **T8: Excessive AI Permissions**

- * Category: Authorization
- * Affected component/data flow: AI File Assistant (P3), LLM Integration Service (P4)

- * Attack description: P3 or P4 gains excessive permissions, compromising file access.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement strict access controls in P2.

****Insecure LLM Context Handling****

1. **T9: Insecure LLM Context Handling**

- * Category: File Security
- * Affected component/data flow: AI File Assistant (P3), Server Storage Repository (DS1)
- * Attack description: An attacker exploits vulnerabilities in context handling, compromising the integrity of files.
- * Impact: Integrity of files is compromised.
- * Severity: Medium
- * Likelihood: Low
- * Mitigation: Implement robust input validation and sanitization in P3.

****External Provider Risks****

1. **T10: External Entity Exploitation**

- * Category: File Security
- * Affected component/data flow: External Entity EE2 (External LLM Provider)
- * Attack description: An attacker exploits vulnerabilities in the external LLM provider, compromising file access.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement robust input validation and sanitization in P4.

****Prompt Injection through Documents****

1. **T11: Prompt Injection**

- * Category: File Security
- * Affected component/data flow: AI File Assistant (P3), Server Storage Repository (DS1)
- * Attack description: An attacker injects malicious prompts into documents, compromising file access.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement robust input validation and sanitization in P3.

****Malicious Document Content****

1. **T12: Malicious Document Content**

- * Category: File Security
- * Affected component/data flow: Server Storage Repository (DS1)
- * Attack description: An attacker embeds malicious content into documents, compromising file access.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Low
- * Mitigation: Implement robust input validation and sanitization in P3.

****Unauthorized File Access****

1. **T13: Unauthorized File Access**

- * Category: Authorization
- * Affected component/data flow: File Management Service (P2), Server Storage Repository (DS1)
- * Attack description: An attacker gains unauthorized access to files.
- * Impact: Confidentiality and Integrity of files are compromised.
- * Severity: High
- * Likelihood: Medium
- * Mitigation: Implement robust authentication and authorization mechanisms in P2.

****Sensitive Document Exposure****

1. **T14: Sensitive Document Exposure**

- * Category: File Security
- * Affected component/data flow: Server Storage Repository (DS1)
- * Attack description: An attacker accesses sensitive documents without proper authorization.
- * Impact: Confidentiality of files is compromised.
- * Severity: High
- * Likelihood: Medium
- * Mitigation: Implement robust encryption and access controls in DS1.

Note that this analysis focuses on the provided architecture and does not assume additional components. The threats listed above are specific to the LLM-integrated secure file management system and may not be exhaustive.